

15-April-2026

Agenda:

Statistics - 8

- Distribution (Normal Dist)
- Zscore
- Z-statistics
- CLT (Central Limit Theorem)

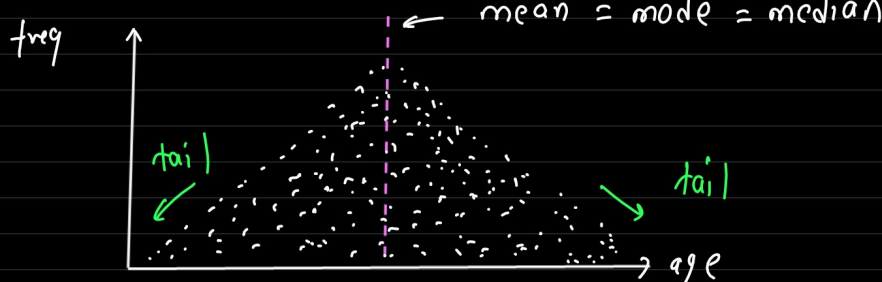
Normal distribution / Gaussian Distribution

- The bell-shaped curve
- Symmetric, continuous, defined for all real number with probability concentrated near the mean.
- It has 2 parameter μ (location), σ (spread)
- In real world normal distribution appears naturally.
 - ↳ Average values happen often
 - ↳ Extreme values happen rarely

Example:

- Human height
- Exam scores
- Avg age

Normal distribution: a model where middle is common & extremes are rare



$$X \sim N(\mu, \sigma^2)$$

$N \rightarrow$ normal
distribution

components:

X : variable

μ : mean

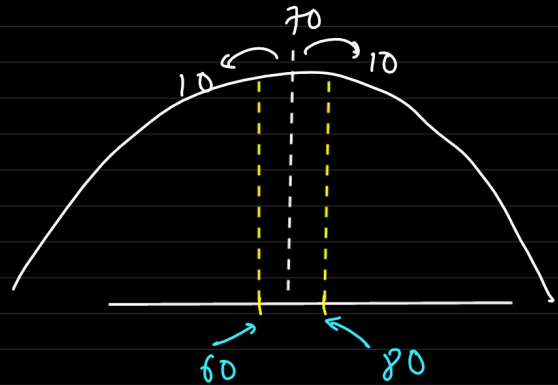
σ : STD (spread)

σ^2 : variance

Scenario:

$$X \sim N(70, 10^2)$$

$X \rightarrow$ marks in exam



Average marks = 70

standard deviation: 10

Question: what is the prob a student scores between 60 & 80?

PMF $\rightarrow X$

PDF $\rightarrow$ continuous value

PDF formula:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$= \int_{60}^{80} \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

lower bound $\rightarrow 60$

upper bound $\rightarrow 80$

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 70}{10} = -1$$

1 std $\leftarrow$ $\rightarrow$ 1 std

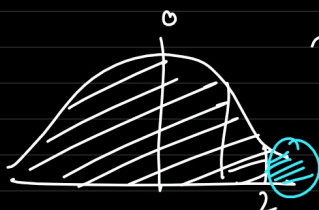
$$p(-1 \leq z \leq 1)$$

$$z = \frac{80 - 70}{10} = 1$$

$$p(60 \leq x \leq 80) = 0.6827$$

Q: student score above 90

$$z = \frac{90 - 70}{10} = 2$$

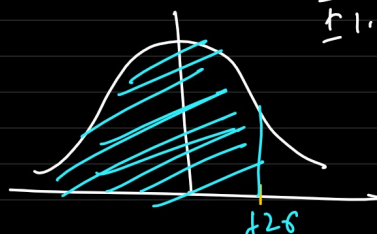
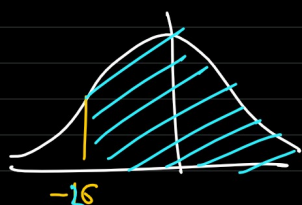


$$p(z > 2) = 0.0228$$

$$(1 - p(z > 2)) = 0.9772$$

$\rightarrow$ prob that student score above 90

$z: 2$



± 1.0
 ± 1.01
 ± 1.02
 ± 1.03

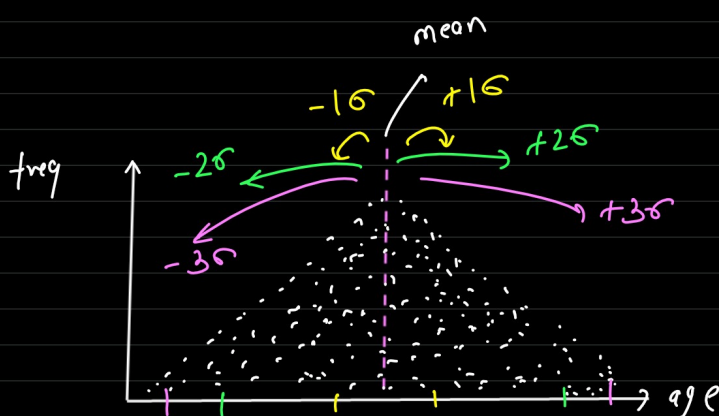
± 2.0
 ± 2.01
 ± 2.03

1 -

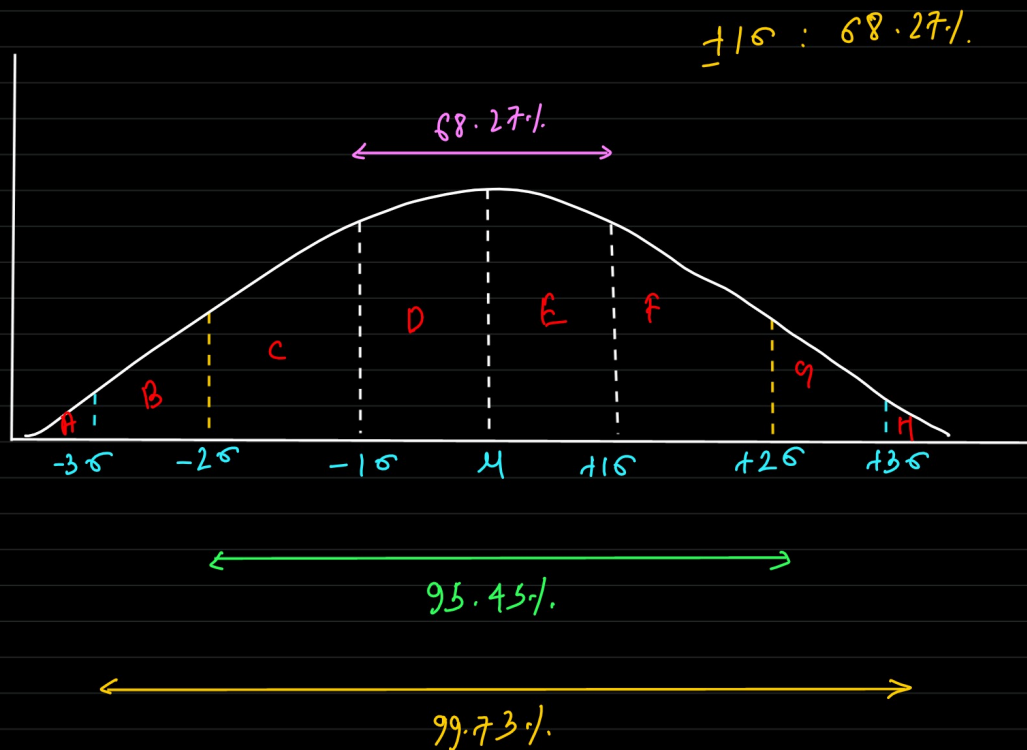
The Empirical Rule (68-95-99.7) simplified explanation

within $\pm 1\sigma$: 68.27%
 within $\pm 2\sigma$: 95.45%
 within $\pm 3\sigma$: 99.73%
 within $\pm 4\sigma$: 99.9937%
 $\pm 5\sigma$: 99.99994%
 $\pm 6\sigma$: 99.9999999%

outside of $\pm 3\sigma$: 0.27% data
 extremely rare events



left tail: ABC
right tail: FGH



in each tail % of value we have ($\pm 1\sigma$):

$$(100 - 68.27) / 2 \rightarrow 15.87\%$$

z-score: How many standard deviation a value is from the mean.

$$z = \frac{x - \mu}{\sigma}$$

$x \rightarrow$ actual value
 $\mu \rightarrow$ mean
 σ : standard deviation : (convert distance into SD units)

class A: dist-1 $\rightarrow$ 100 students $\rightarrow$ student_id $\rightarrow$ 15 $\rightarrow$ 60 $\rightarrow$ $\xrightarrow{65}$ $\xrightarrow{2/100}$

class B: dist-2 $\rightarrow$ 100 students $\rightarrow$ student_id $\rightarrow$ 18

: 90/100 $\rightarrow$ topper
 $\downarrow$
 80

60
 $\downarrow$
 last-rank

Raw distance alone can mislead:

Exam A mean = 70, SD = 5

Exam B mean = 70, SD = 20

A score of 80 is A: 10

B: 10

in A: 2 SD away

in B: 0.5 SD away

} SD give more context

full example:

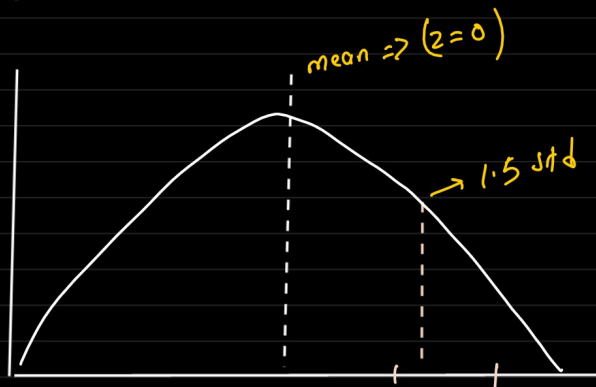
Mean = 70

SD = 10

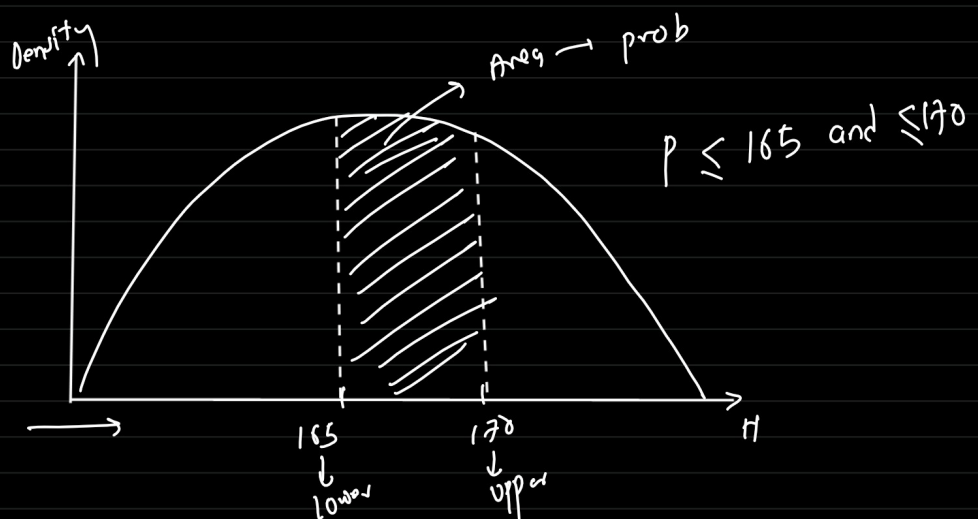
Student scored: $x = 85$

find z-score:

$$\frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5 \text{ above average}$$



pdf →



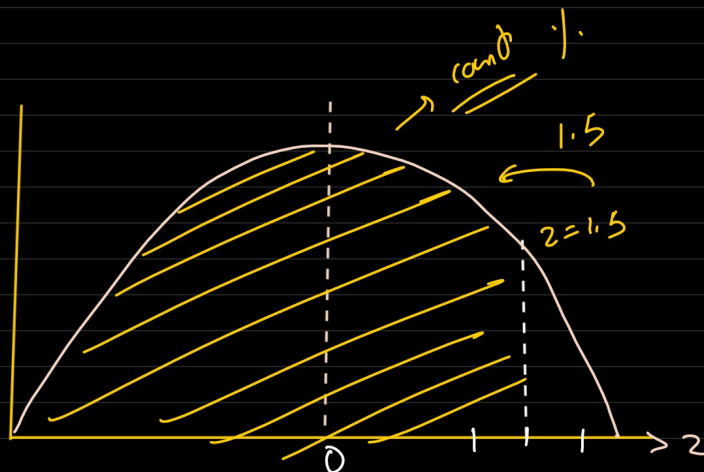
pdf $\rightarrow$

Density

prob of student height
 > 180

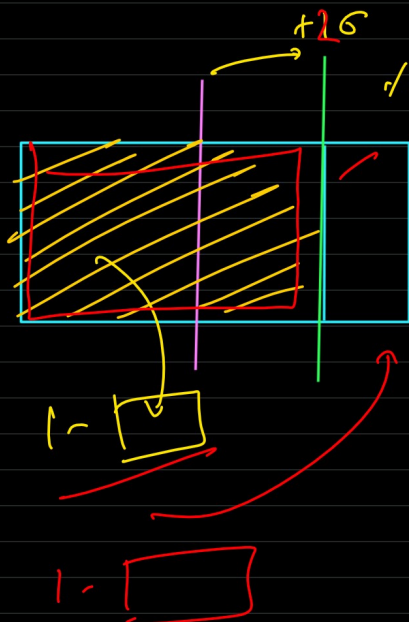
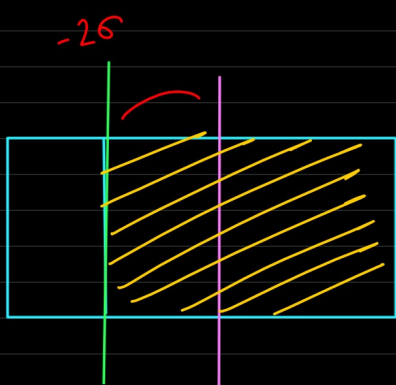


$\mu = 150$
 $\downarrow$
 z value



vald $\rightarrow 2$
 1000 $\rightarrow 2$

$z = 1.5$



$\square$

$$2 \rightarrow \frac{1.68}{0.08} \quad 1.6 \quad \boxed{\frac{0.08}{0.95352}}$$

$$-2.94 \rightarrow -2.9 \quad \boxed{\frac{0.09}{0.00164}} \downarrow \underline{\underline{\text{prob.}}}$$

11-12th